Cover letter

25 August 2026

The Editors
Chaos, Solitons & Fractals

Dear Editors,

We submit **"Forgeable greenbeards: multistability and hollow collapse in certified agent populations"** for consideration as a regular paper in *Chaos, Solitons & Fractals*. The manuscript is authored by Trung-Kiet Huynh, Dao-Sy Duy-Minh, Chi-Nguyen Tran, Pham Phu Hoa, and Nguyen Lam Phu Quy.

The paper studies a 48-strategy replicator flow on a 47-dimensional simplex, together with its finite-population counterpart. Each strategy carries a genuine, forged or absent identity tag and conditions its conduct on whether a partner's tag passes verification. The forged-tag pass rate turns the classical greenbeard vulnerability into a family of bifurcation thresholds.

The main contribution is to nonlinear population dynamics. Transversal eigenvalues yield the principal invasion boundaries in closed form, while numerical integration characterizes basin structure and a separate targeted rest-face robustness probe. The two invasion boundaries meet at a codimension-two point: increasing assortment removes the visible exploiter while leaving a behavioural mimic essentially unchanged. Uniformly sampled replicator trajectories split between disjoint certified and uncertified safe rest faces, whereas targeted starts reach a numerically robust unsafe mixed rest face outside that sample. This multistability shows why evaluating a nonlinear population observable at a basin-weighted mean state can create interactions absent from every underlying attractor.

The analysis also connects the dynamics to certification design. A reciprocating club tolerates more than twice the forgery of an unconditionally trusting club; the fine required for marginal protection diverges as forgery becomes undetectable; and behavioural mimics can destroy credential integrity before conduct deteriorates. Certification therefore buys robustness to displacement rather than a lower harm level within either dominant settled regime, and the exclusionary conduct that sustains this robustness imposes an entry barrier on compliant unbadged agents.

The manuscript fits the journal's scope through nonlinear dynamics, self-organization and an application to social and engineered multi-agent systems. It emphasizes qualitative changes in the flow, is written for an interdisciplinary audience, and uses numerical computation to support rather than replace the analytical results.

This manuscript has not been peer reviewed or formally published and is not under consideration elsewhere. A non-peer-reviewed development-source version has been publicly accessible on GitHub since 19 August 2026; it is a preprint, not a version of record. Two related but non-duplicate companion manuscripts by overlapping author teams are separate submissions in the same coordinated round: "Delegation cascades in evolutionary games: stable computation with strategic depth" and "Safe designs, unsafe ecosystems: deployment-layer selection in an evolutionary AI race." All three use the same cited, unchanged four-strategy interaction layer as a common benchmark, but introduce distinct strategic variables, state spaces, analyses and results. Copies can be supplied for confidential editorial assessment if requested.

The code and machine-readable results that regenerate every quoted scalar, table and figure are publicly available at https://github.com/trungkiet2005/greenbeard-tag. All five authors approved the manuscript, its author order and this submission, and declare no competing interests.

Thank you for your consideration.

Yours sincerely,

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